

1. Title

Case Study on Microsoft Azure: Deployment of a Static Web Application

2. Problem Statement

Microsoft Azure is a cloud computing platform and infrastructure created by Microsoft for building, deploying, and managing applications and services through a global network of Microsoft-managed data centers.

The objective of this case study is to understand Azure services and deploy a static website using Azure cloud infrastructure.

3. Objectives

- To understand the concept of **cloud computing**
 - To study the features of Microsoft Azure
 - To learn how to deploy a **static web application**
 - To gain practical experience with Azure portal
 - To understand cloud-based hosting and scalability
-

4. Introduction to Microsoft Azure

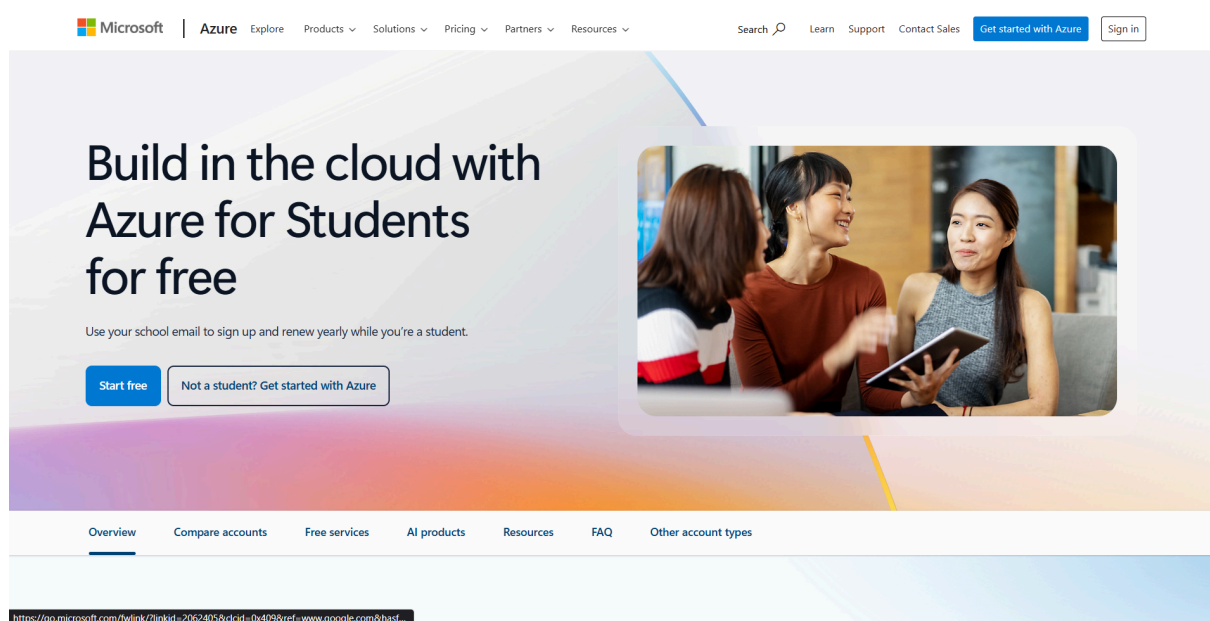
Microsoft Azure is a cloud computing platform that provides services such as:

- Virtual Machines
- Storage
- Networking
- Databases
- AI & Analytics
- Web Hosting

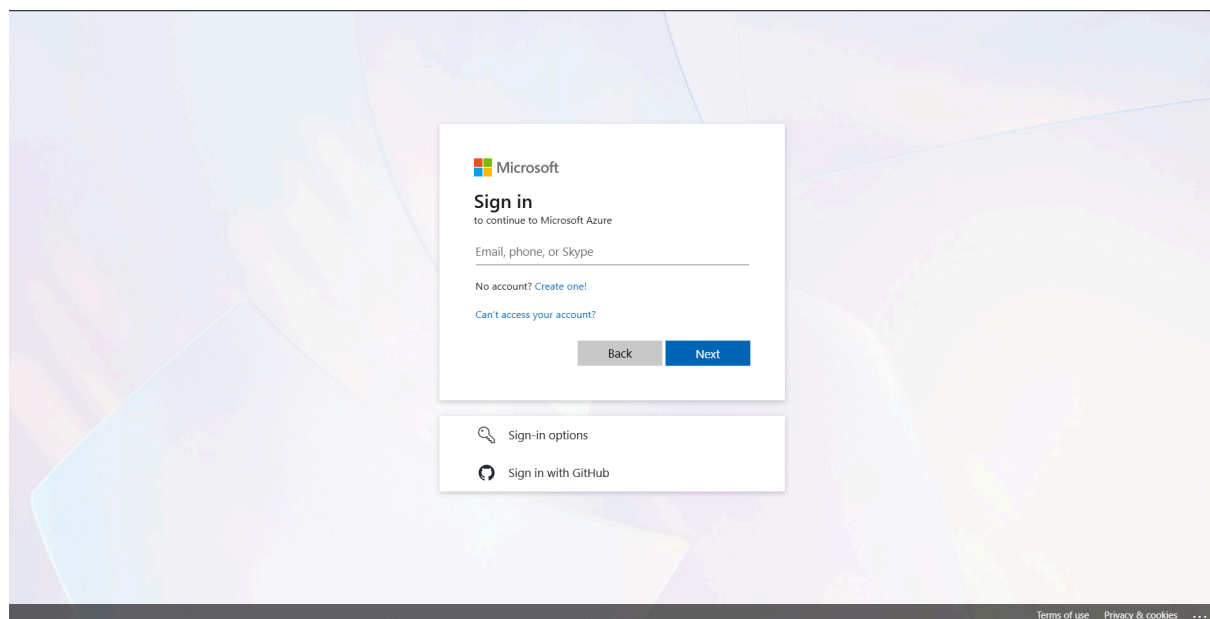
Steps to Perform: Hosting Static Website on Azure

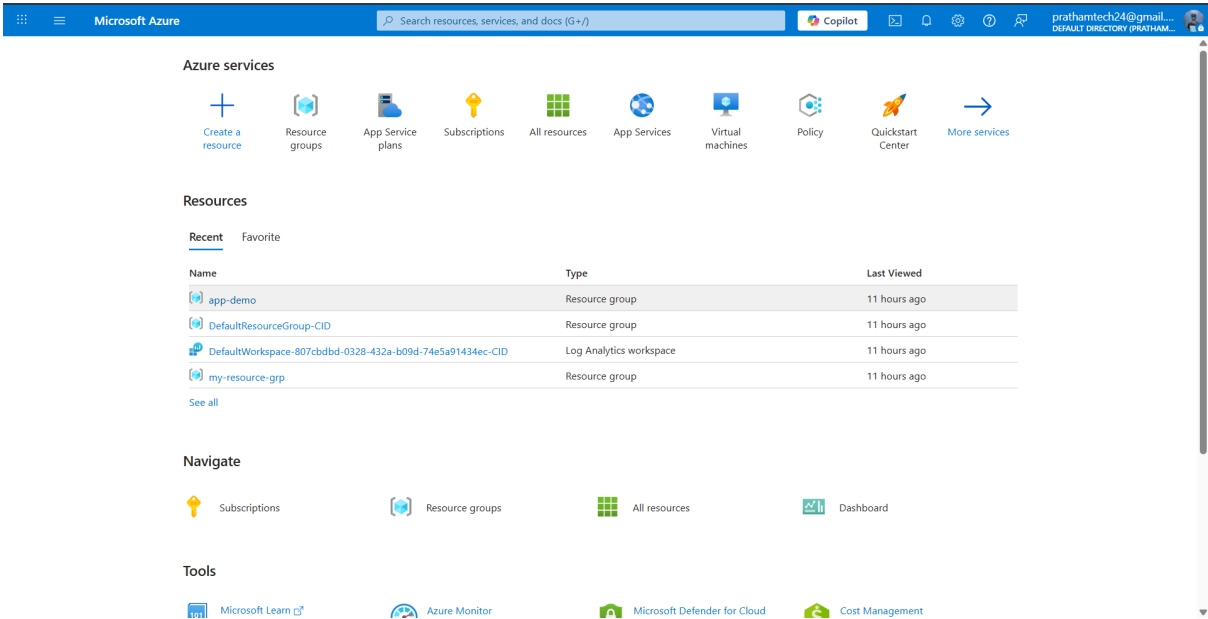
Deployment Method: Azure Static Web Apps (GitHub Integration)

1. Account Setup and Authentication

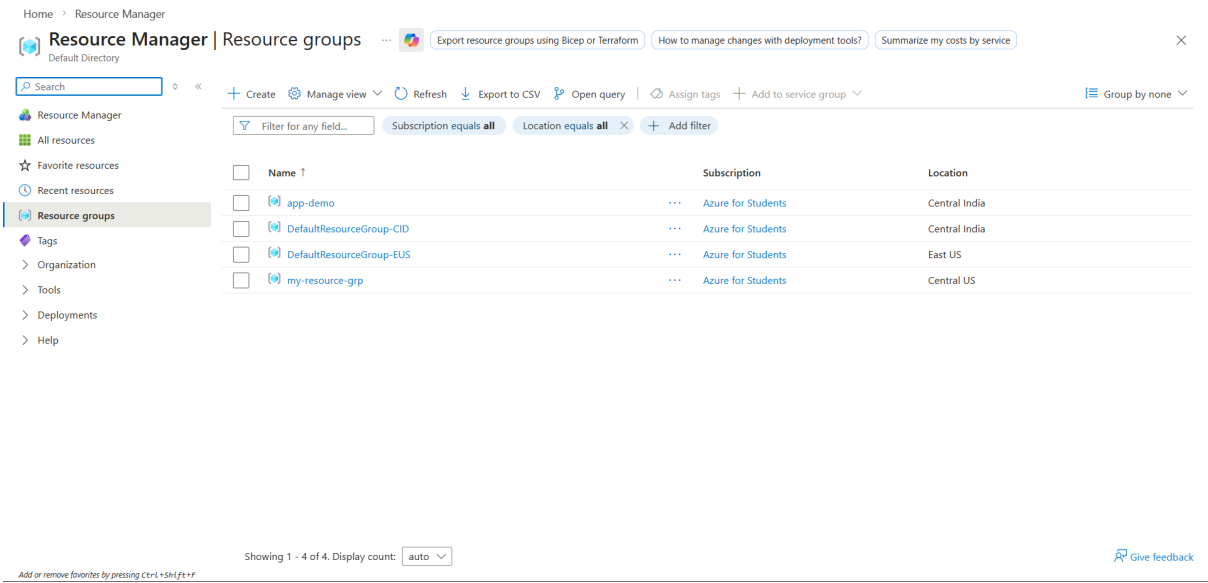


2. Step: Created an **Azure for Students** account.





3. Resource Group (RG) Creation:



Microsoft Azure

Search resources, services, and docs (G+)

Copilot

prathamtech24@gmail...
DEFAULT DIRECTORY (PRATHAM...

Home > Resource Manager | Resource groups

Create a resource group

BasicsTagsReview + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * Azure for Students

Resource group name * Azure-hosting-demo

Region * (Asia Pacific) Central India

Previous

Next

Review + create

App Service Plan: Search App Service Plans Create.

Microsoft Azure

Search resources, services, and docs (G+)

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Create a resource

App Service plans

App Services

Virtual machines

Policy

Quickstart Center

More services

Resources

RecentFavorite

Name	Last Viewed
app-demo	11 hours ago
DefaultResourceGroup-CID	11 hours ago
DefaultWorkspace-807cbdbd-0328-432a-b09d-74e5a91434ec-CID	11 hours ago
my-resource-grp	11 hours ago

See all

Navigate

SubscriptionsResource groupsAll resourcesDashboard

Tools

Microsoft Defender for CloudCost Management

App Service plans

CreateView

Description

App Service plans represent the collection of physical resources used to host your apps, like location, scale, size and SKU

Learn more with Copilot

https://portal.azure.com/#blade/HubsExtension/BrowseResourceBlade/resourceType/Microsoft.Web%2Fserverfarms

Microsoft Defender for CloudCost Management

Microsoft Azure

Search resources, services, and docs (G+)

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Home > App Service plans

Create App Service Plan

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource Group * Azure-hosting-demo

App Service Plan details

Name * Azure-hosting-demo-plan

Operating System * Linux Windows

Region * Central India

Pricing Tier

App Service plan pricing tier determines the location, features, cost and compute resources associated with your app. [Learn more](#)

Pricing plan Free F1 (Shared infrastructure)

Zone redundancy

Review + create < Previous Next : Tags >

Search Static Web Apps

Microsoft Azure

Search resources, services, and docs (G+)

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Home > Create a resource

Marketplace

Get Started

Service Providers

Search with AI

Management

Private Marketplace

Private Offer Management

My Marketplace

Favorites

My solutions

Recently created

Private plans

Categories

Infrastructure Services (53)

Web (7)

IT & Management Tools (3)

Developer Tools (2)

static web app

Pricing : All Operating System : All Publisher Type : All Product Type : All Publisher name : All

☐ Azure benefit eligible only ☐ Azure services only

New! Get AI-generated suggestions for 'static web app' [View suggestions](#)

Showing 1 to 20 of 64 results for 'static web app'. [Clear search](#)

Static Web App

Microsoft

Azure Service

Enjoy secure and flexible development, deployment, and scaling options for your web app

Create

Websoft9 Web Runtime for Python

VMLAB INC.

Virtual Machine

Pre-configured, web-based, cloud-native, secure, one-click to deploy Python App with Websoft9 Applications Hosting Platform on

Starts at ₹3,279/hour

Create

MoxVeda - Website & App Localization Platform

Process Nine Technologies Pvt. L...

SaaS

A no-code platform to seamlessly translate your website, app, documents, chatbots & more

Starts at ₹1,817,563.65/3 years

Subscribe

Nextjs

Cloud Infrastructure Services

Virtual Machine

Next.js On Ubuntu 24.04. React framework that enables developers to build fast, scalable, and SEO-friendly web applications. Optimise

Starts at ₹4,198/3 years

Create

Nginx Proxy Server

Cloud Infrastructure Services

Virtual Machine

Nginx on Windows Server 2022 - Use for proxy, web server, reverse proxy, caching, load balancing or mail proxy.

Starts at ₹3,157/hour

Create

Is Marketplace helpful?

Create.

The screenshot shows the Microsoft Azure Marketplace page for the 'Static Web App' service. The header includes the Microsoft Azure logo, a search bar, and user information. The main content area displays the 'Static Web App' card with a 3.8 rating and 113 reviews. Below the card, there are tabs for 'Overview', 'Plans', 'Usage Information + Support', and 'Ratings + Reviews'. The 'Overview' tab is selected, showing a description of the service and a list of other products from Microsoft. The 'Create' button is visible at the bottom right of the card.

Home > Create a resource > Marketplace

Static Web App

Microsoft

Static Web App [Add to Favorites](#)

Microsoft | Azure Service

★ 3.8 (113 ratings)

Subscription: Azure for Students Plan: Static Web App [Create](#)

Overview Plans Usage Information + Support Ratings + Reviews

With Static Web Apps, you can automatically build and deploy full-stack web apps to Azure from a code repository. Your static content—like HTML, CSS, JavaScript, and images—is globally distributed from points around the world instead of a traditional web server, so files are closer to end users. You can even include integrated serverless APIs using Azure Functions or leverage Server Side Rendering.

More products from Microsoft [See All](#)

- Active Directory Health Check**
Microsoft
Azure Service
Assess the risk and health of Active Directory.
- AD Replication Status**
Microsoft
Azure Service
Identify Active Directory replication issues in your environment.
- Device Update for IoT Hub**
Microsoft
Azure Service
Securely and Reliably update your devices with Device Update for IoT.
- Front Door and CDN profiles**
Microsoft
Azure Service
Azure Front Door and CDN profiles is security led, modern cloud CDN that

<https://portal.azure.com/#> [Give feedback](#)

Basics Tab:

- **Subscription:** Azure for Students.
- **Resource Group:** Select Azure-hosting-demo.
- **Name:** React-hosting (or your project name).
- **Plan Type:** Free (For hobby/student projects).

The screenshot shows the 'Create Static Web App' configuration page in the Microsoft Azure portal. The page is divided into sections for 'Project Details', 'Hosting region', 'Static Web App details', and 'Hosting plan'. The 'Project Details' section shows the subscription as 'Azure for Students' and the resource group as 'Azure-hosting-demo'. The 'Hosting region' section shows the region as 'Global'. The 'Static Web App details' section shows the name as 'React-hosting'. The 'Hosting plan' section shows the plan type as 'Free: For hobby or personal projects'. The 'Next: Deployment configuration >' button is visible at the bottom right.

Home > Create a resource > Marketplace > Static Web App

Create Static Web App

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource Group * [Create new](#)

Hosting region

Static Web Apps distributes your app's static assets globally. Configure regional features in [Advanced](#)

Regions: Global

Static Web App details

Name *

Hosting plan

The hosting plan dictates your bandwidth, custom domain, storage, and other available features. [Compare plans](#)

Plan type

☒ Free: For hobby or personal projects

☐ Standard: For general purpose production apps

[Review + create](#) [Previous](#) [Next: Deployment configuration >](#)

Select **GitHub**. Sign in with **GitHub**.

Authorize Azure to access your repositories.

Select your **Organization**, **Repository**, and **Branch** (usually main).

The screenshot shows the 'Create Static Web App' page in the Microsoft Azure portal. The page is titled 'Create Static Web App' and has a breadcrumb trail: Home > Create a resource > Marketplace > Static Web App. The 'Deployment details' section is active, showing the 'Source' as 'GitHub'. The 'GitHub account' is 'Pratham-ghadge', with a 'Change account' button. A blue information box states: 'If you can't find an organization or repository, you might need to enable additional permissions on GitHub. You must have write access to your chosen repository to deploy with GitHub Actions.' Below this, the 'Organization' is 'Pratham-ghadge', the 'Repository' is 'CLOUDX', and the 'Branch' is 'main'. The 'Build Details' section is also visible, showing 'Build Presets' as 'React (detected)' and 'App location' as '/'. At the bottom, there are buttons for 'Review + create', '< Previous', and 'Next : Deployment configuration >'.

The screenshot shows the 'Create Static Web App' page in the Microsoft Azure portal, specifically the 'Deployment configuration' section. The page has a breadcrumb trail: Home > Static Web Apps. The 'Deployment configuration' section is active, showing the 'Deployment authorization policy' as 'GitHub'. At the bottom, there are buttons for 'Review + create', '< Previous', and 'Next : Advanced >'.

Advanced Networking:

- Check **Allowed Region Policy** to ensure compliance with Azure's student sandbox restrictions.

The screenshot shows the 'Create Static Web App' wizard in the Microsoft Azure portal, specifically the 'Advanced' tab. The breadcrumb trail is 'Home > Static Web Apps'. The page title is 'Create Static Web App'. The tabs are 'Basics', 'Deployment configuration', 'Advanced' (selected), 'Tags', and 'Review + create'. A paragraph explains that Azure Static Web Apps provide global high availability and managed Azure Functions API. Below this, the 'Azure Functions and staging details' section shows a dropdown menu for 'Region for Azure Functions API and staging environments' set to 'East Asia'. The 'Enterprise-grade edge' section has a description and an unchecked checkbox. A blue information box states: 'Enterprise-grade edge requires a standard hosting plan. [Change hosting plan](#)'. At the bottom, there are three buttons: 'Review + create' (highlighted in blue), '< Previous', and 'Next : Tags >'.

Finalize: Click Review + Create.

The screenshot shows the 'Create Static Web App' wizard in the Microsoft Azure portal, specifically the 'Review + create' tab. The breadcrumb trail is 'Home > Static Web Apps'. The page title is 'Create Static Web App'. The tabs are 'Basics', 'Deployment configuration', 'Advanced', 'Tags', and 'Review + create' (selected). The 'Summary' section shows the 'Static Web App' icon and 'by Microsoft'. The 'Details' section is a table with the following information:

Subscription	1ad08183-06fa-4fe0-a018-10a632436912
Resource Group	Azure-Static-Web-App2
Name	cloudx
Region	eastasia
SKU	Free
Repository	https://github.com/Pratham-ghadge/CLOUDX
Branch	main
App location	/
API location	
Output location	build
Deployment authorization policy	GitHub

At the bottom, there are three buttons: 'Create' (highlighted in blue), '< Previous', and 'Next >'. A link 'Download a template for automation' is also present.

Phase 4: Automation & Deployment (The "Trigger")

Once created, Azure automatically injects a workflow file into your GitHub repository.

1. **GitHub Actions:** Go to your GitHub Repository **Actions** tab.
2. **Workflow Execution:** You will see a live "Job" running. This is the **CI/CD pipeline** compiling your code and pushing it to Azure servers.
3. **Deployment Success:** Once the green checkmark appears in GitHub, return to the Azure Portal.

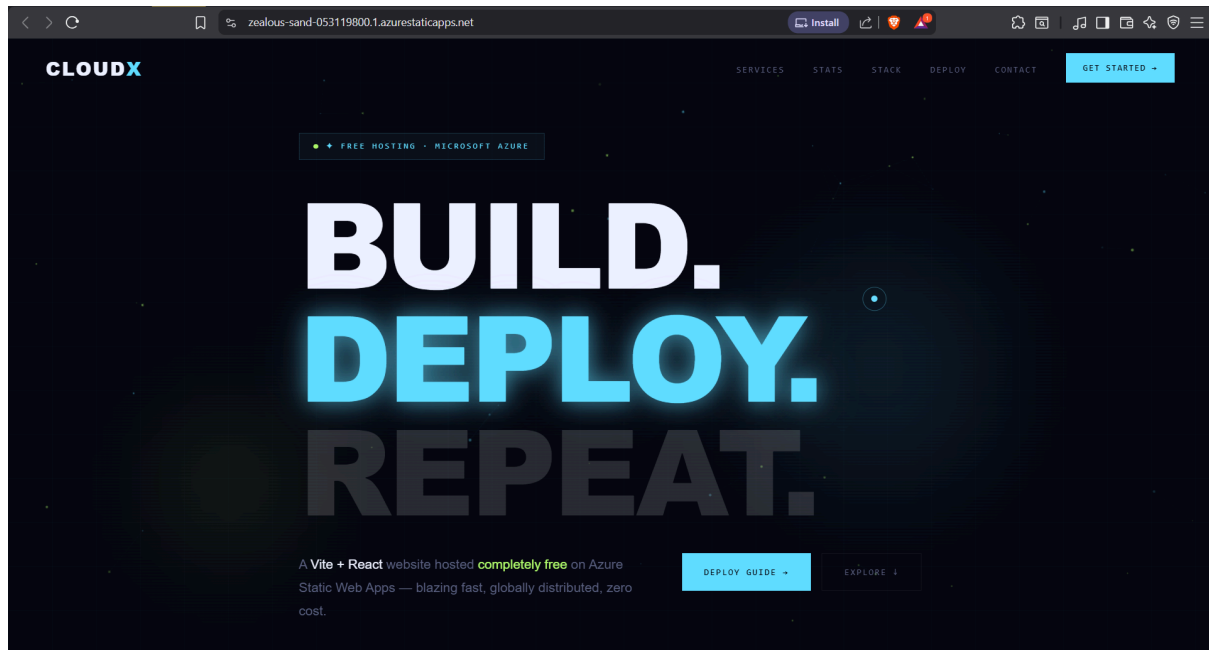
Once Azure displays the "Deployment Succeeded" message, follow these steps to verify the cloud environment.

1. **Resource Navigation:** Click the **Go to resource** button to open the Static Web App Management dashboard.

The screenshot shows the Azure portal interface. At the top, there's a search bar and navigation icons. The main content area displays the 'Overview' tab for a specific deployment. A green checkmark indicates the deployment is complete. Below this, deployment details are listed, including the name, subscription, resource group, start time, and correlation ID. A 'Go to resource' button is prominently displayed. On the right side, there's a sidebar with various links and notifications, including a 'Deployment succeeded' message and links to cost management, security, and tutorials.

Live Access: Click the URL link. This will open your website in a new browser tab.

This screenshot shows the 'cloudx' Static Web App overview in the Azure portal. The interface includes a search bar and navigation icons. The 'Overview' tab is selected, showing a list of links for managing the app. The 'Essentials' section provides key information about the app, including the resource group, subscription, subscription ID, location, sku, and tags. A 'View your application' section displays the status, environment, domain, and hosting plan. A 'Visit your site' button is available. At the bottom, there are three cards for 'Add a custom domain', 'Upgrade your hosting plan', and 'Enable enterprise grade edge'.



Conclusion :

By utilizing the **Azure Student Developer Pack**, this project successfully demonstrates a modern **DevOps workflow**. By decoupling the frontend hosting from a traditional server and using a **CI/CD pipeline**, we ensure that the deployment process is bug-free, automated, and highly scalable for future enhancements.